



COL7378: Principles of Autonomous Systems

Semester II, 2025-26

Sate Estimation - I

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Today's lecture

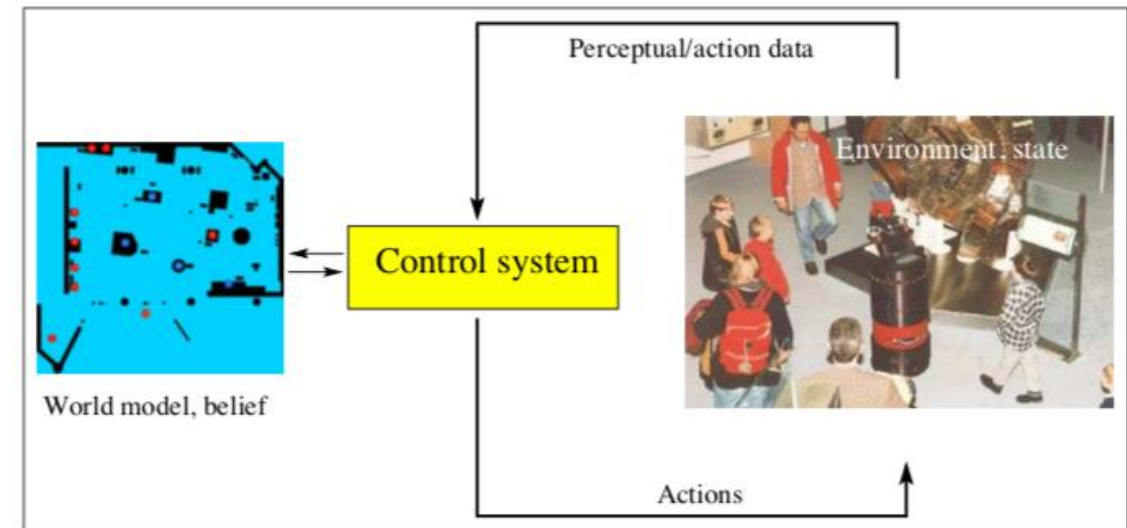
- Last Class
 - Policy Gradients
- This Class
 - State Estimation
 - Recursive State Estimation
 - Bayes Filter
 - References
 - Probabilistic Robotics Ch 1 & 2
 - AIMA Ch 15 (till sec 15.3)

Acknowledgements

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Robot Environment Interaction

- Environment or world
 - Objects, robot, people, interactions
 - Environment possesses a true internal state
- Observations
 - The agent cannot directly access the true environment state.
 - Takes observations via its sensors which are error prone.
- Belief
 - Agent maintains a belief or an estimate with respect to the state of the environment derived from observations.
 - The belief is used for decision making
- Actions
 - Agent can influence the environment through its physical interactions (actuators, motions, language interaction etc.)
 - The effect of actions may be stochastic.
 - Taking actions affects the world state and the robot's internal belief with regard to this state.

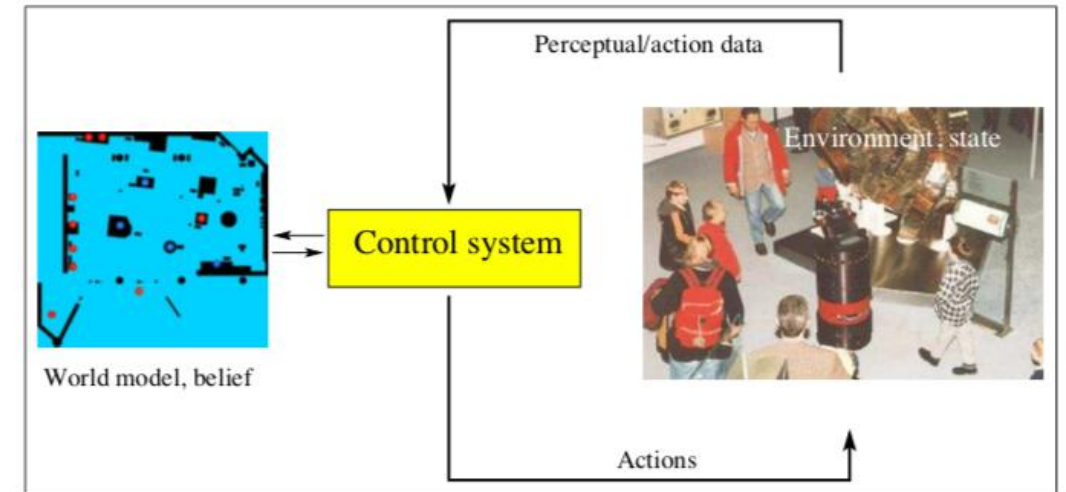


State Estimation

- Framework for estimating the state from sensor data.
- Estimating quantities that are *not directly observable*.
 - *But* can be inferred if certain quantities are available to the agent.
- State estimation algorithms
 - Compute *belief distributions over possible states* of the world.

State

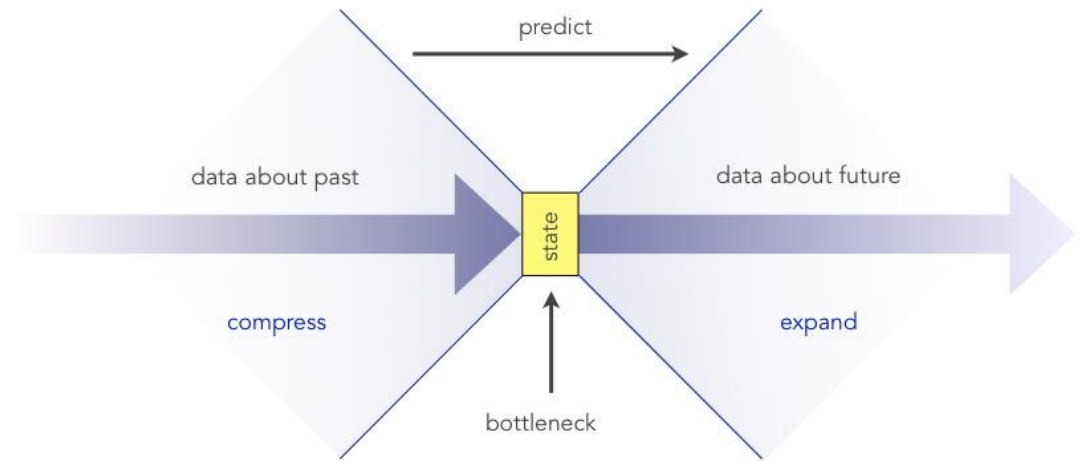
- What is typically part of the state, x ?
 - Robot pose: position and orientation or kinematic state
 - Velocities: of the robot and other objects like people.
 - Location and features of surrounding objects in the environment.
 - Semantic states: is the door open or closed?
 -
- What is put in the state is influenced by which task we seek to perform
 - Navigation
 - More complex example (e.g., delivery of hospital supplies)



State

- Environment is characterized by the state.
 - *“A collection of all aspects of the agent and its environment that can impact the future”*
- A sufficient statistic of the past observations and interactions required for future tasks.
- State plays an important role for decision making.

Figure courtesy Byron Boots



State: statistic of history sufficient to predict the future

Markovian assumption:

Future is independent of past given present

Two aspects: Sensing and Taking Actions

- **Taking Sensor Measurements**

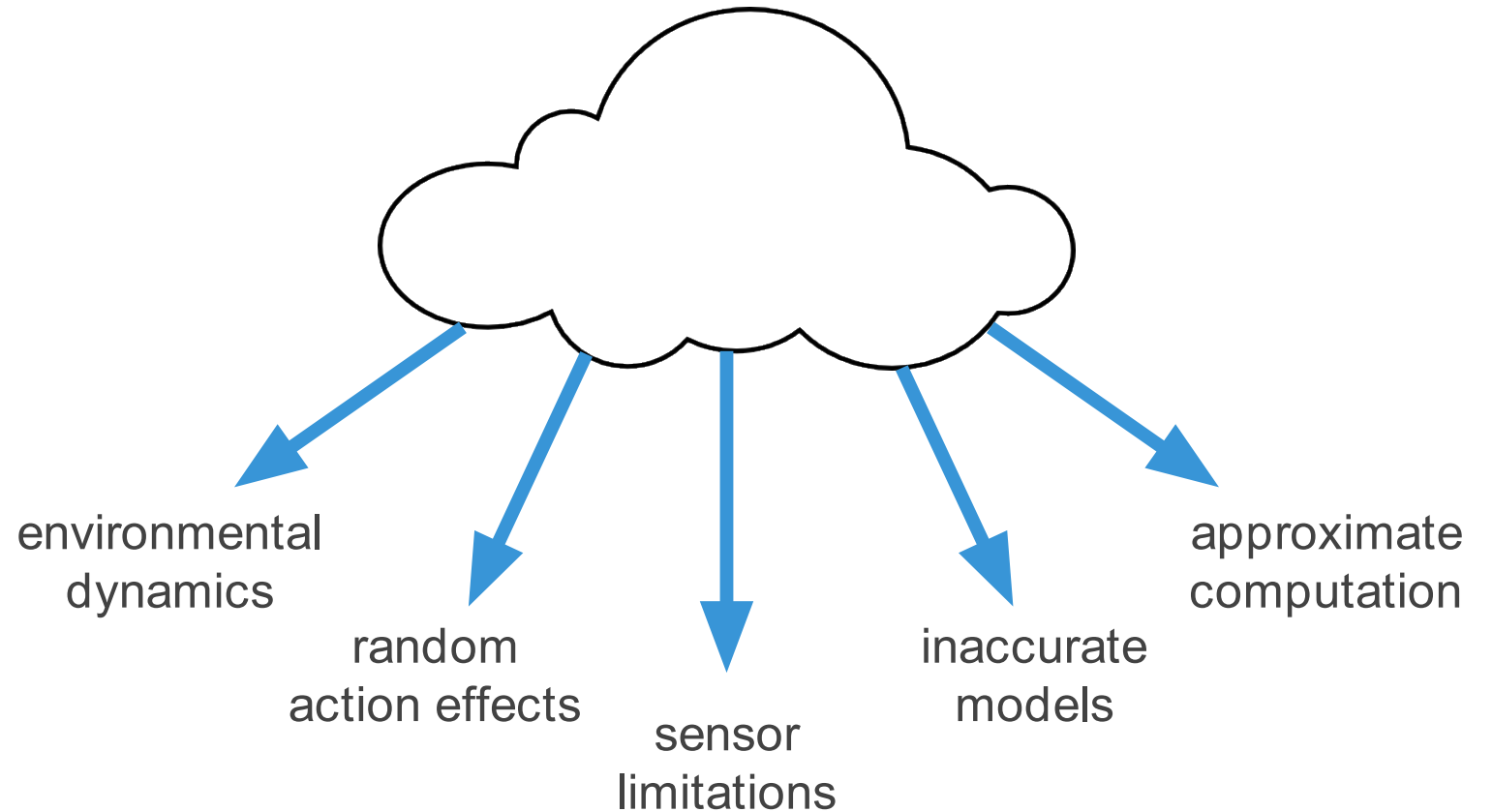
- Camera, range, tactile, language query etc.
- Denote measurement data as z_t
- Noisy observations of the true state.
- Measurements typically add information, *decrease uncertainty*.

- **Taking Actions (or Controls)**

- Physical interaction: robot motion, manipulation of objects, *NO_OP* etc.
- Carry information about the change of state.
- Source of control data: odometers or wheel encoders.
- Denote control data as u_t
- Actions are never carried out with absolute certainty.
- In contrast to measurements, actions generally *increase uncertainty*.

Uncertainty

Explicitly represent uncertainty using probability theory.



Probability Recap

Independence

- X and Y are **independent** iff

$$P(x,y) = P(x) P(y)$$

- $P(x \mid y)$ is the probability of **x given y**

$$P(x \mid y) = P(x,y) / P(y)$$

$$P(x,y) = P(x \mid y) P(y)$$

- If X and Y are **independent** then

$$P(x \mid y) = P(x)$$

Marginalization

Discrete case

$$\sum_x P(x) = 1$$

$$P(x) = \sum_y P(x, y)$$

$$P(x) = \sum_y P(x \mid y) P(y)$$

Continuous case

$$\int p(x) dx = 1$$

$$p(x) = \int p(x, y) dy$$

$$p(x) = \int p(x \mid y) p(y) dy$$

Conditioning

- Law of total probability:

Marginalize out z .

$$P(x) = \int P(x, z) dz$$

$$P(x) = \int P(x | z) P(z) dz$$

Conditioning on the extra variables.

$$P(x | y) = \int P(x | y, z) P(z | y) dz$$

Conditional Independence

- X and Y are conditionally independent given Z.
- Given Z , X does not add information about Y and vice versa.

$$P(x, y | z) = P(x | z)P(y | z)$$

$$P(x | z) = P(x | z, y)$$

$$P(y | z) = P(y | z, x)$$

Bayes Rule

$$P(x, y) = P(x | y)P(y) = P(y | x)P(x)$$

$\Rightarrow$

$$P(x | y) = \frac{P(y | x) P(x)}{P(y)} = \frac{\text{likelihood} \cdot \text{prior}}{\text{evidence}}$$

$$P(x | y) = \frac{P(y | x) P(x)}{P(y)} = \eta P(y | x) P(x)$$

$$\eta = P(y)^{-1} = \frac{1}{\sum_x P(y | x) P(x)}$$

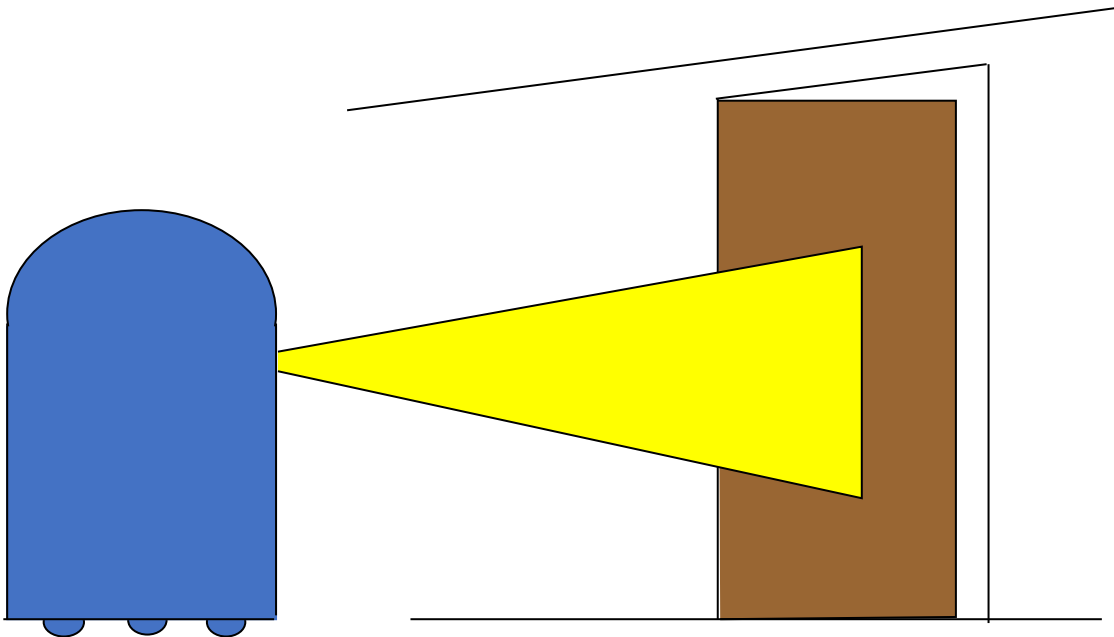
Bayes Rule with Background Knowledge

When extra information is available. Incorporate that knowledge as observations of extra variables.

$$P(x \mid y, z) = \frac{P(y \mid x, z) P(x \mid z)}{P(y \mid z)}$$

Example of State Estimation

- The robot wants to estimate the state of the door as closed or open
 - Has a noisy sensor that produces measurement, z
- Estimate: $P(\text{open} | z)$?
 - Likelihood that the true state of the door is open given that z was measured.



$$P(z | \text{open}) = 0.6 \quad P(z | \neg \text{open}) = 0.3$$

$$P(\text{open}) = P(\neg \text{open}) = 0.5$$

The observation z is correlated with the true state as open or not-open.

E.g., measuring a particular distance or classifying an image.

Causal vs. Diagnostic Reasoning

- $P(\text{open} | z)$ is diagnostic reasoning
 - *Given that I observe z what is the likelihood that the door state is actually open?*
- $P(z | \text{open})$ is causal reasoning (can estimate by counting frequencies)
 - *Given that the door state is open what is the likelihood of getting measurement z ?*
- Often causal knowledge is easier to obtain
 - Given the underlying state collect the data.
- Bayes rule enables the use of causal knowledge:

$$P(\text{open} | z) = \frac{P(z | \text{open})P(\text{open})}{P(z)}$$

Incorporating a single measurement

- **Higher** likelihood of observation z when the door **is open** compared to when the door is closed.
- The incorporation of the measurement z **raises** the probability that the door is open.

$$P(z \mid open) = 0.6 \quad P(z \mid \neg open) = 0.3$$

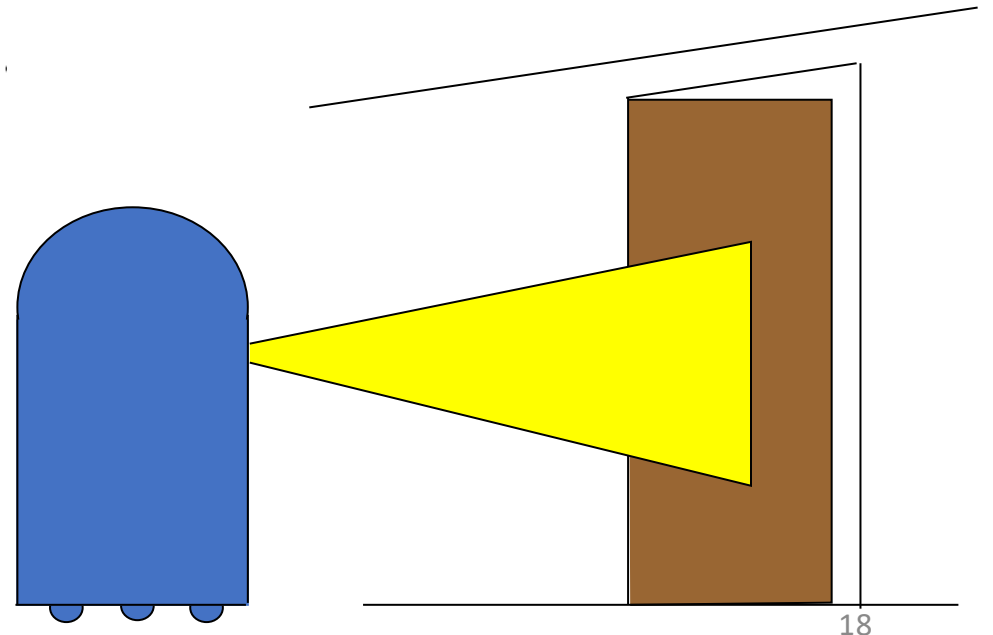
$$P(open) = P(\neg open) = 0.5$$

$$P(open \mid z) = \frac{P(z \mid open)P(open)}{P(z \mid open)P(open) + P(z \mid \neg open)P(\neg open)}$$

$$P(open \mid z) = \frac{0.6 \cdot 0.5}{0.6 \cdot 0.5 + 0.3 \cdot 0.5} = \frac{2}{3} = 0.67$$

Incorporating multiple measurements

- Suppose the robot has another sensor that produces a second observation z_2
- How can we combine the measurement of the second sensor
- What is $P(\text{open} | z_1, z_2)$?
- In general, how to estimate $P(x | z_1 \dots z_n)$



Recursive Bayesian Updating

$$P(x \mid z_1, \dots, z_n) = \frac{P(z_n \mid x, z_1, \dots, z_{n-1}) P(x \mid z_1, \dots, z_{n-1})}{P(z_n \mid z_1, \dots, z_{n-1})}$$

Markov assumption: z_n is conditionally independent of $z_1, \dots, z_{n-1}$ given x .

$$\begin{aligned} P(x \mid z_1, \dots, z_n) &= \frac{P(z_n \mid x) P(x \mid z_1, \dots, z_{n-1})}{P(z_n \mid z_1, \dots, z_{n-1})} \\ &= \eta P(z_n \mid x) P(x \mid z_1, \dots, z_{n-1}) \\ &= \eta_{1\dots n} \prod_{i=1\dots n} P(z_i \mid x) P(x) \end{aligned}$$

In our causal modeling view, the world state is *causing* all the observations.

Incorporating second sensor measurement

- Higher likelihood of observation z when the door is **not** open compared to when the door is open.
- The inclusion of the second measurement z_2 **lowers** the probability for the door to be open.

$$\begin{aligned} P(z_2 \mid open) &= 0.5 & P(z_2 \mid \neg open) &= 0.6 \\ P(open \mid z_1) &= 2/3 & P(\neg open \mid z_1) &= 1/3 \end{aligned}$$

$$\begin{aligned} P(open \mid z_2, z_1) &= \frac{P(z_2 \mid open) P(open \mid z_1)}{P(z_2 \mid open) P(open \mid z_1) + P(z_2 \mid \neg open) P(\neg open \mid z_1)} \\ &= \frac{\frac{1}{2} \cdot \frac{2}{3}}{\frac{1}{2} \cdot \frac{2}{3} + \frac{1}{5} \cdot \frac{1}{3}} = \frac{5}{8} = 0.625 \end{aligned}$$

Sensor Model

- Sensor model
 - What is the likelihood of obtaining this sensor measurement given the true state?
 - A conditional distribution over observations given the true state. Generative Model.
 - Observations or measurements can be considered as the noisy projection of the state

$$p(z_t | x_t)$$

Action Model

- Action or Motion model
 - How the actions or controls change the state of the world?
 - Incorporate the outcome of an action u into the current “belief”, we use the conditional distribution.
 - Specifies how does the state change by application of the action (from the state, x_{t-1} to the state, x_t by executing the action, u_t).

$$p(x_t | x_{t-1}, u_t)$$

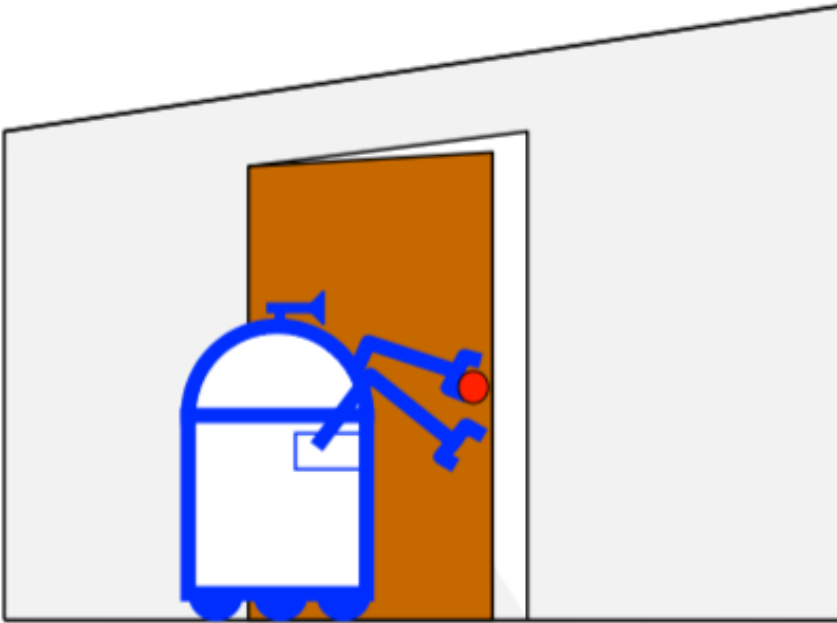
Belief over the world state

- Belief
 - Expresses the agent's internal knowledge about the state of an aspect of the world.
 - Note: we do not know the true state.
 - The belief is estimated from the sensor measurement data and the actions taken till now.

$$Bel(x_t) = p(x_t | z_{1:t}, u_{1:t})$$

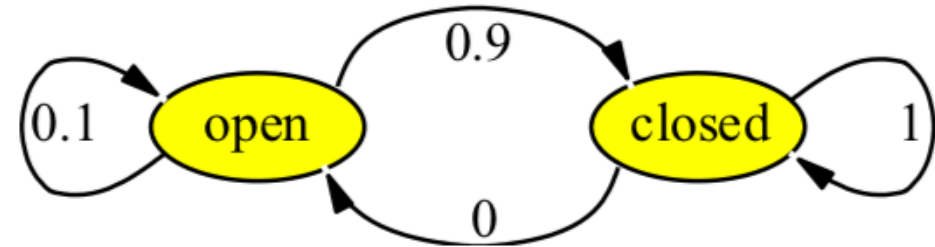
Example: Incorporating action effects

Example: Closing the door



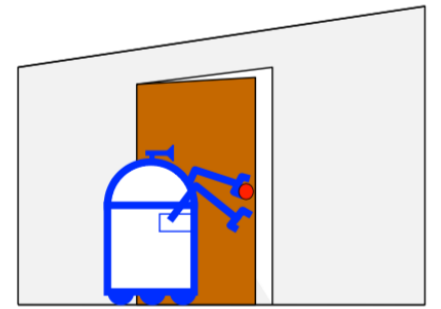
Probabilistic effects

$P(x|u, x')$ for $u = \text{"close door"}$:



If the door is open, the action “close door” succeeds in 90% of all cases.

Example: The Resulting Belief



Marginalizing out the outcome of actions using the distribution over the state at the last time step.

Continuous case:

$$P(x | u) = \int P(x | u, x') P(x') dx'$$

Discrete case:

$$P(x | u) = \sum P(x | u, x') P(x')$$

$$\begin{aligned} P(\text{closed} | u) &= \sum P(\text{closed} | u, x') P(x') \\ &= P(\text{closed} | u, \text{open}) P(\text{open}) \\ &\quad + P(\text{closed} | u, \text{closed}) P(\text{closed}) \\ &= \frac{9}{10} * \frac{5}{8} + \frac{1}{1} * \frac{3}{8} = \frac{15}{16} \\ P(\text{open} | u) &= \sum P(\text{open} | u, x') P(x') \\ &= P(\text{open} | u, \text{open}) P(\text{open}) \\ &\quad + P(\text{open} | u, \text{closed}) P(\text{closed}) \\ &= \frac{1}{10} * \frac{5}{8} + \frac{0}{1} * \frac{3}{8} = \frac{1}{16} \\ &= 1 - P(\text{closed} | u) \end{aligned}$$

Incorporating Measurements

- Bayes rule

$$P(x|z) = \frac{P(z|x)P(x)}{P(z)} = \frac{\text{likelihood} \cdot \text{prior}}{\text{evidence}}$$

Bayes Filter

- **Given:**

- Stream of observations z and action data u :
- Sensor model
- Action model
- Prior probability of the system state $P(x)$.

$$d_t = \{u_1, z_2 \dots, u_{t-1}, z_t\}$$

$$p(z_t | x_t)$$

$$p(x_t | x_{t-1}, u_t)$$

- **What we want to estimate?**

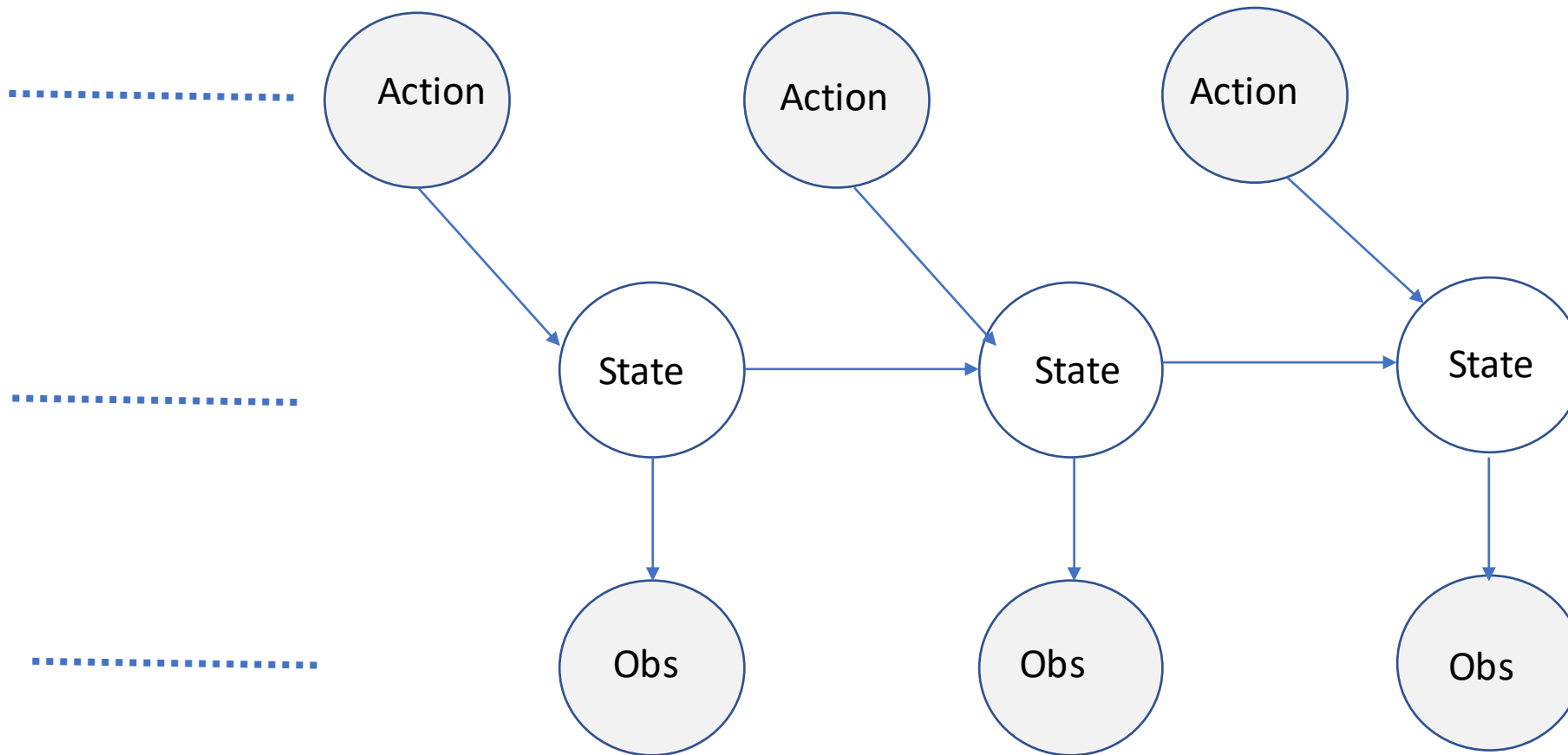
- The state at time t
- A belief or the posterior over the state:

$$Bel(x_t) = P(x_t | u_1, z_2 \dots, u_{t-1}, z_t)$$

Intuitively: Given all observations collected by the agent till time t and all the actions taken by the agent till time t , what is our estimate over its state?

In essence

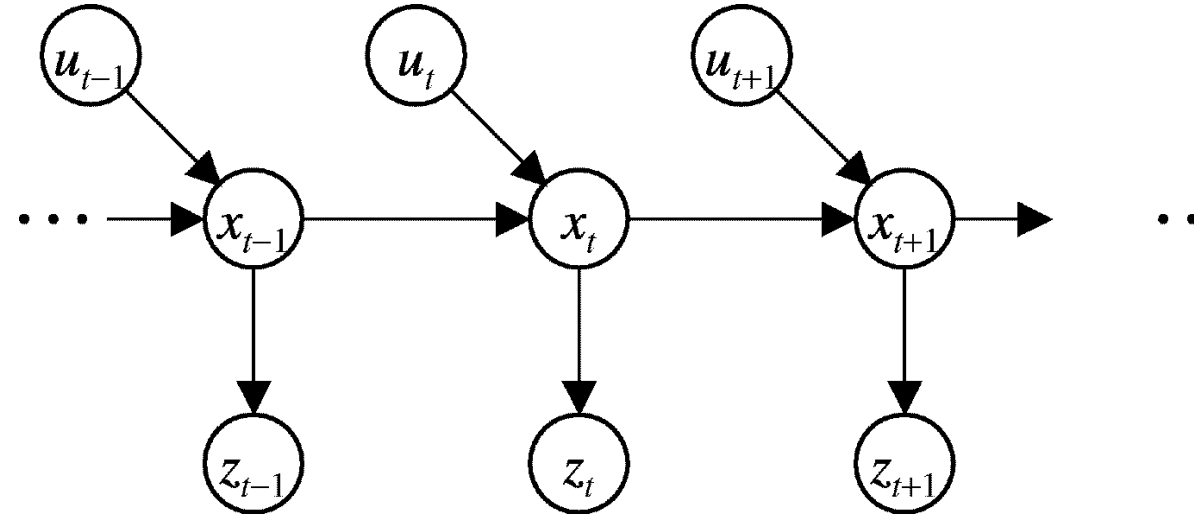
We estimate the state of the agent via measurements and knowledge of what actions were taken. Bayes Filter provides a recursive way to estimate the likelihood given the conditional independence assumptions.



Formally, a Generative Model

Assumptions

- Static world
- Independent noise
- Perfect model, no approximation errors
- Markov assumption (once you know the state the past actions and observations do not affect the future).

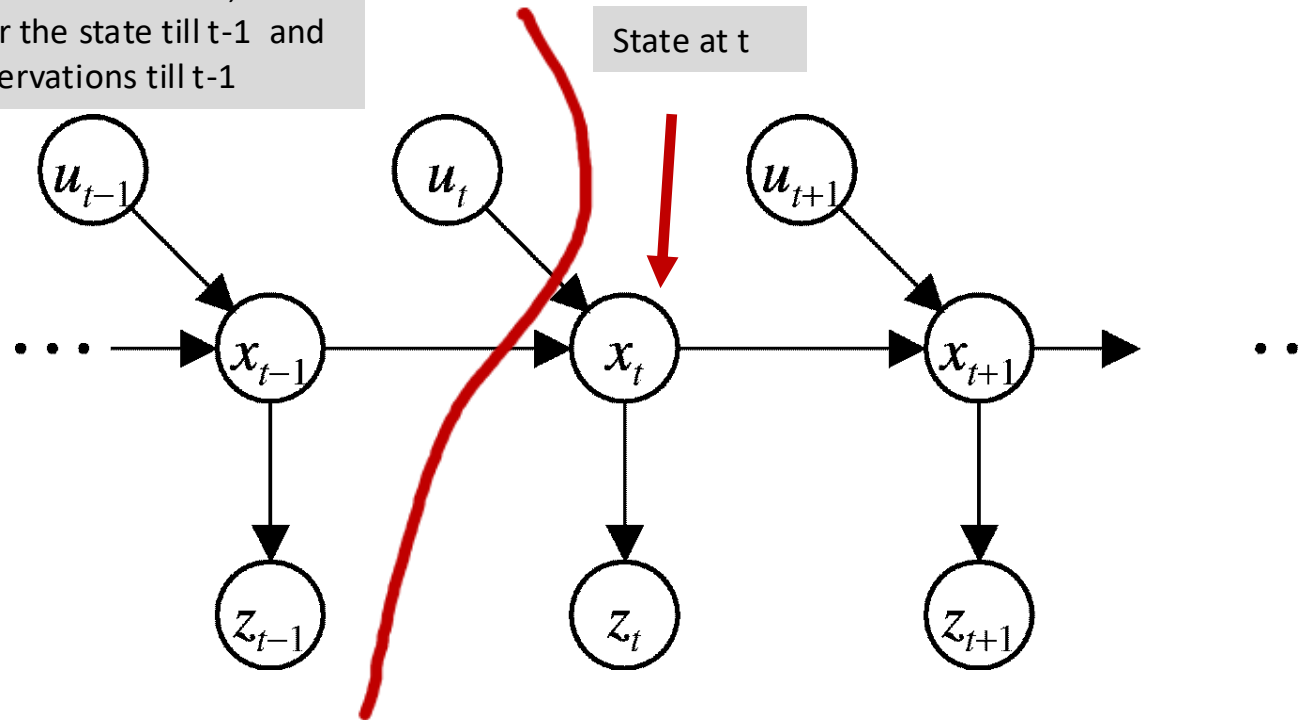


$$p(z_t \mid x_{0:t}, z_{1:t-1}, u_{1:t}) = p(z_t \mid x_t)$$

$$p(x_t \mid x_{1:t-1}, z_{1:t-1}, u_{1:t}) = p(x_t \mid x_{t-1}, u_t)$$

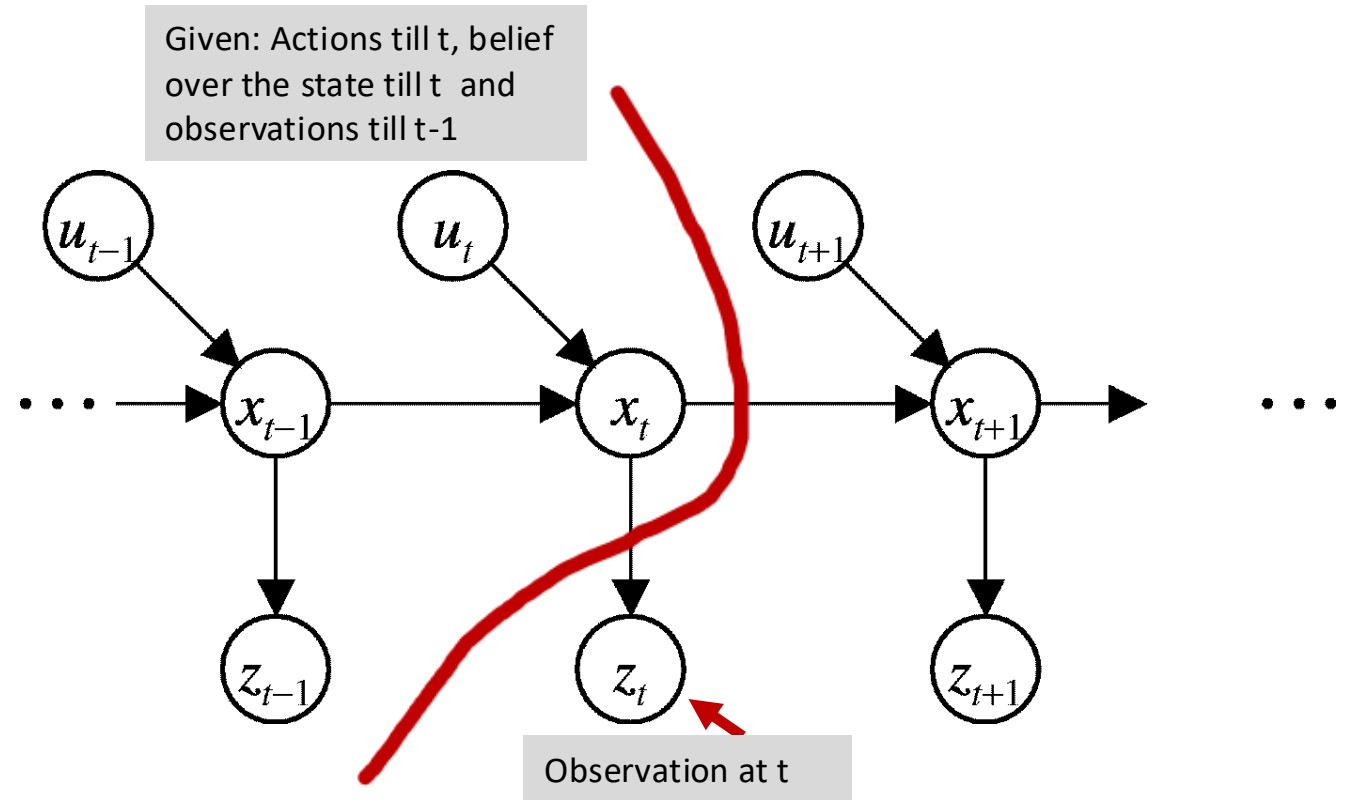
Generative Model

Given: Actions till t, belief over the state till t-1 and observations till t-1



$$p(x_t \mid x_{1:t-1}, z_{1:t-1}, u_{1:t}) = p(x_t \mid x_{t-1}, u_t)$$

Generative Model



$$p(z_t \mid x_{0:t}, z_{1:t-1}, u_{1:t}) = p(z_t \mid x_t)$$

Bayes Filters

z = observation
 u = action
 x = state

$$\boxed{Bel(x_t)} = P(x_t \mid u_1, z_1, \dots, u_t, z_t)$$

Bayes
$$= \eta P(z_t \mid x_t, u_1, z_1, \dots, u_t) P(x_t \mid u_1, z_1, \dots, u_t)$$

Markov
$$= \eta P(z_t \mid x_t) P(x_t \mid u_1, z_1, \dots, u_t)$$

Total prob.

$$= \eta P(z_t \mid x_t) \int P(x_t \mid u_1, z_1, \dots, u_t, x_{t-1}) P(x_{t-1} \mid u_1, z_1, \dots, u_t) dx_{t-1}$$

Markov
$$= \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) P(x_{t-1} \mid u_1, z_1, \dots, u_t) dx_{t-1}$$

$$\boxed{= \eta P(z_t \mid x_t) \int P(x_t \mid u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}}$$

Bayes Filters Algorithm

1. **Algorithm Bayes_filter** ($Bel(x), d$):
2. $n=0$
3. **If** d **is a perceptual data item** z **then**
4. For all x do
5. $Bel'(x) = P(z | x) Bel(x)$
6. $\eta = \eta + Bel'(x)$
7. For all x do
8. $Bel'(x) = \eta^{-1} Bel'(x)$
9. **Else if** d **is an action data item** u **then**
10. For all x do
11. $Bel'(x) = \int P(x | u, x') Bel(x') dx'$
12. **Return** $Bel'(x)$

$Bel(x)$ is a distribution of
belief over states

$$Bel(x_t) = \eta P(z_t | x_t) \int P(x_t | u_t, x_{t-1}) Bel(x_{t-1}) dx_{t-1}$$

Bayes Filter: Takeaways

- Bayes filters are a probabilistic tool for estimating the state of with observations acquired over time.
 - Target is to obtain the belief over the current state given past actions and observations.
 - Estimate this distribution in a recursive manner.
 - Update using actions
 - Update using measurements.
- Bayes rule allows us to compute probabilities that are difficult to determine otherwise.
- Under the Markov assumption, recursive Bayesian updating can be used to efficiently combine evidence.